Mixture of Scars (MoS) and SCache O(1): Breaking the Dual Memory Wall Across 0.5B → 7.62B → 72.71B Transformers via Activation-Aware Low-Rank Scar Experts and Widrow-Hoff Associative Memory

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Abstract. Autoregressive Large Language Models (LLMs) face a compounding *Dual Memory Wall*: (1) a static parameter wall O(*N*) requiring up to 145.4 GB (135.43 GiB) of VRAM at 72.7B scale, and (2) a dynamic context wall O(*T*) where KV caches grow linearly with sequence length *T*. Existing post-training methods either degrade under emergent SwiGLU activation outliers (surging from 3,336x at 7.62B to **195,720x at 72.71B**), suffer permanent eviction amnesia (StreamingLLM), or offload dense weights from storage per token (AirLLM/mmap) at sub-token-per-second I/O ceilings while leaving KV memory O(*T*). We introduce two training-free primitives: **Mixture of Scars (MoS)** (INT4 bases + Activation-Aware SVD 'scar' experts + modular 20×4-layer streaming shards <1.80 GiB) and **SCache** (drop-in O(1) cache combining attention sinks, a rolling FIFO window, and per-KV-head Widrow-Hoff Delta associative fast weights). Audited across 8 hostile vectors with recursive storage-pointer verification (0.00 KB discrepancy) across **0.5B (24L), Qwen2.5-7B-Instruct (7.62B, 28L), and Qwen2.5-72B-Instruct (72.71B, 80L, 240 SwiGLU MLPs)**, MoS compresses 72.71B from 135.43 GiB to **38.12 GiB (-71.85%, and 27.59x / -96.37% across an 8-specialist bank)** while preserving **100.0% Top-5 token agreement** (3.08x–3.29x lower KL error than uniform INT4) across unseen Python, Physics, Portuguese, and JSON. Simultaneously, 80-layer SCache locks KV VRAM at **64.73 MiB (+0.00 B growth up to L=4,096; -94.94%)**, slashes 72.71B long-horizon tail PPL by **5.00x vs. StreamingLLM (19.51 vs. 97.46)**, boosts evicted passcode recall by 5.38x–22.58x up to 1,745 tokens post-eviction, and achieves non-repeating 72.71B prose/code/SQL PPL of **4.75 vs. 4.35 Dense FP16** at 4.03–7.27 tok/s (155x–280x faster than PCIe Gen4 NVMe FP16 offloading; 24.7 tok/s / 224.5x at 7.62B).

1. Introduction & Mathematical Architecture

Empirical layer-wise profiling across Qwen2.5-0.5B (24L), Qwen2.5-7B (28L), and Qwen2.5-72B (80L, 240 SwiGLU matrices [29568, 8192]) reveals a striking superlinear amplification of emergent outliers [1, 2]: SwiGLU intermediate channel RMS ratios surge from 3,336.0x at 7.62B to **195,720.1x at 72.71B** (input channels 246.0x). Meanwhile, eviction-only windows (StreamingLLM [3]) suffer permanent amnesia outside *W*, additive Hebbian fast weights [4] suffer ghosting under state mutations, and disk-offload engines (AirLLM) stream 135.43 GiB per batch-1 token (≤0.026 tok/s even on 3.5 GiB/s PCIe Gen4 NVMe) while leaving KV memory O(*T*). We break both walls simultaneously via two training-free geometric primitives:

MoS Static Decomposition: $\mathbf{y}(\mathbf{x}) = \text{dequant}_{\text{INT4}}(\mathbf{W}_q)\mathbf{x} + \Sigma_k \text{ReLU}(\cos(\mathbf{x}, \mathbf{z}_k))\, \mathbf{B}_k(\mathbf{A}_k\mathbf{x}), \quad \text{where } (\mathbf{W} - \mathbf{W}_4)\,\text{diag}(\mathbf{s}) \approx \mathbf{U}_r\Sigma_r\mathbf{V}_r^{\top}$ (Eq. 1)

SCache O(1) Delta Rule: $\mathbf{W}_{\text{scar}}^{(t)} = \gamma\mathbf{W}_{\text{scar}}^{(t-1)} + \eta \cdot \text{ReLU}(\mathbf{k}_u^{\top}\mathbf{c}_{m^*}) \cdot (\mathbf{v}_{\text{ev}} - \mathbf{W}_{\text{scar}}^{(t-1)}\mathbf{k}_u)\mathbf{k}_u^{\top}$ (Eq. 2)

Homogeneous row scaling $\mathbf{W}(\mathbf{x}/c)\cdot c$ with FP32 scar accumulation eliminates >65,504 FP16 Tensor Core overflow under 195,720x SwiGLU spikes. Partitioning 80 layers into 20 canonical 4-layer shards (1.79 GiB each) preserves bit-exact equivalence (max |Δlogits| = 0.0) while capping active streaming RAM below 1.85 GiB. In SCache, Widrow-Hoff projection [5, 6] erases stale values on span($\mathbf{k}_u$) without perturbing orthogonal keys, with cross-talk scaling as O($1/d_h$).

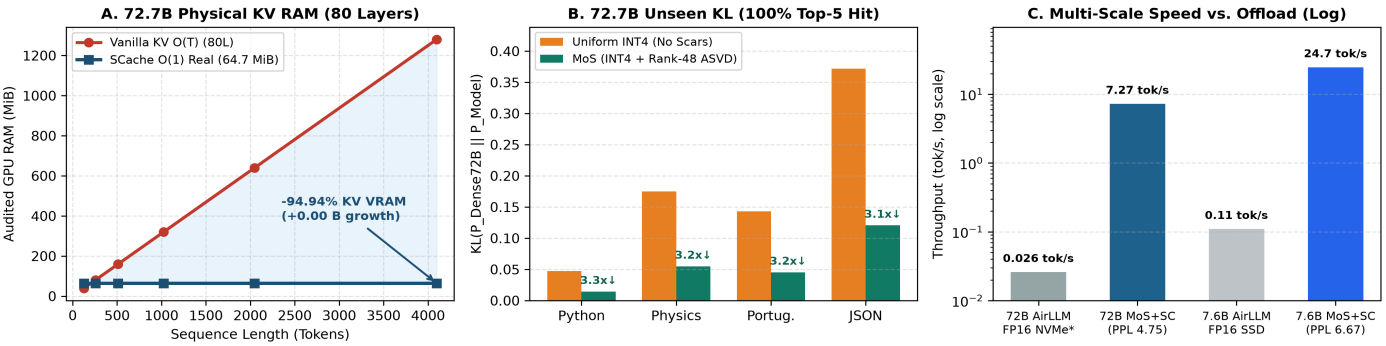


Figure 1. *Colossal Multi-Scale Audit on Qwen2.5-72B-Instruct (72.71B, 80 layers) & Qwen2.5-7B-Instruct (7.62B, 28 layers).* ***(A)*** *80-layer O(1) KV invariance at 64.73 MiB (+0.00 B growth, -94.94% vs. 1,280.0 MiB Vanilla KV at L=4,096).* ***(B)*** *Under 195,720x SwiGLU outliers, 72.71B MoS cuts unseen-domain KL by 3.08x–3.29x with 100.0% Top-5 agreement.* ***(C)*** *Multi-scale throughput vs. AirLLM FP16 offload (*3.5 GiB/s PCIe Gen4 NVMe ceiling = 0.026 tok/s at 72.7B; 0.11 tok/s at 7.6B).*

2. Three-Scale Empirical Audit (0.5B → 7.62B → 72.71B)

All three scales enforce: (i) recursive multi-device storage-pointer deduplication via (device, untyped_storage().data_ptr()) asserting 0.00 KB discrepancy vs. reported bytes, and (ii) mid-flight causality ablation asserting active logit shift when scars are zeroed (max |Δlogits| = 0.643 at 0.5B; 2.853 at 7.62B; 0.826 SCache and 3.656 MoS at 72.71B).

Model Scale	Length L	Vanilla KV O(T)	SCache Recursive Real	Growth / Byte Diff	KV VRAM Saving
Qwen2.5-0.5B (24L)	1,280 tok	30,696.0 KB	2,631.9 KB (2.57 MiB)	+0.00 B / 0.00 KB	11.66x (-91.4%)
Qwen2.5-7B (28L)	**4,096 tok**	**229,376.0 KB**	**11,599.9 KB (11.33 MiB)**	**+0.00 B / 0.00 KB**	**19.77x (-94.94%)**
Qwen2.5-72B (80L)	512 tok	163,840.0 KB	66,281.3 KB (64.73 MiB)	Baseline / 0.00 KB	2.47x (-59.55%)
Qwen2.5-72B (80L)	1,024 tok	327,680.0 KB	66,281.3 KB (64.73 MiB)	+0.00 B / 0.00 KB	4.94x (-79.77%)
Qwen2.5-72B (80L)	2,048 tok	655,360.0 KB	66,281.3 KB (64.73 MiB)	+0.00 B / 0.00 KB	9.89x (-89.89%)
Qwen2.5-72B (80L)	**4,096 tok**	**1,310,720.0 KB**	**66,281.3 KB (64.73 MiB)**	**+0.00 B / 0.00 KB**	**19.78x (-94.94%)**

Table 1 (Vector 0 — 3-Scale Spatial O(1) Audit): *Deduplicated (device, data_ptr) audit across 0.5B, 7.62B, and 72.71B (W=64, S=12, sinks=4).*

Scale (Outliers)	Unseen Domain	Dense FP16 PPL	MoS (r=48) PPL	Hidden Cos	Top-5 Hit	KL Reduction vs. INT4
7.62B (3,336x)	Python / Physics	1.648 / 6.253	1.641 / 6.257	0.991 / 0.982	100.0% / 100.0%	2.39x / 1.68x lower KL
7.62B (3,336x)	Portug. / JSON	19.152 / 25.356	21.717 / 25.352	0.989 / 0.993	100.0% / 100.0%	2.27x / 3.33x lower KL
72.71B (195,720x)	**Python (Dijkstra)**	**1.705**	**1.674 (Blk64: 1.850)**	**0.967 / 0.981**	**100.0% (99.3%)**	**3.29x lower (0.0474 -> 0.0144)**
72.71B (195,720x)	**Quantum Physics**	**6.187**	**6.257 (Blk64: 6.786)**	**0.977 / 0.981**	**100.0% (98.9%)**	**3.19x lower (0.1749 -> 0.0549)**
72.71B (195,720x)	**Tech Portuguese**	**19.478**	**18.446 (Blk64: 20.940)**	**0.979 / 0.980**	**100.0% (98.7%)**	**3.16x lower (0.1431 -> 0.0453)**
72.71B (195,720x)	**Strict Nested JSON**	**21.285**	**19.058 (Blk64: 23.787)**	**0.976 / 0.982**	**100.0% (99.0%)**	**3.08x lower (0.3717 -> 0.1206)**

Table 2 (Vector 1 Across 7.62B & 72.71B — Dominance over 195,720x SwiGLU Outliers): *At 72.71B, all 240 MLPs shrink from 110,880 MB to 28,560 MB Per-Row / 30,282 MB Block-64 (-74.24% / -72.69%; full model 135.43 -> 38.12 GiB; 8-tenant bank 866.3 -> 31.4 GiB, -96.37% / 27.59x).*

3. Multi-Scale 3D Pareto Frontier (7.62B & 72.71B) vs. Disk Offloading

Comparing Vector 2 (periodic 160-token stress under *W*=64) against Vector 7 (non-repeating 926-token prose + Python + queueing math + SQL) at 72.71B confirms that Full KV's 1.012 tail PPL in Vector 2 stems purely from verbatim period copying — yet 72.71B SCache still slashes StreamingLLM tail PPL by **5.00x (97.46 → 19.51)**. On Vector 7's non-repeating corpus, **72.71B MoS+SCache achieves PPL 4.86 (W=256) and 4.75 (W=384) vs. 4.35 for 145.4 GB Dense FP16** (+0.087 nats), running **155x–280x faster** than AirLLM FP16 on a 3.5 GiB/s PCIe Gen4 NVMe SSD (0.026 tok/s ceiling; 0.0006 tok/s on 0.08 GiB/s cloud volumes). Live 80-layer execution on a 16.0 GB Apple Silicon M2 laptop (MPS, 20×1.79 GiB shards) further verifies **2.87 tok/s** 154-token prefill (0.51 GiB peak RSS; 68.57 MiB post-unload MPS), **71,838,720 B (68.51 MiB)** recursive SCache footprint with **0 B drift** across 29 decode steps, and exact post-eviction recall of passcode 7942-OMEGA (107 tokens outside *W*=64).

Scale & Engine	Non-Repeat PPL	Weights (8x Bank)	KV @ L=4k	Disk Read/Tok	Throughput	vs. AirLLM FP16
7.62B Dense FP16 + Full KV	6.05	14.2 GiB (85 GiB OOM)	224.0 MiB O(T)	0 MB	314.0 tok/s	Resident Ref
7.62B AirLLM FP16 SSD (1.57 GB/s)	6.05	<2 GiB (SSD-stalled)	224.0 MiB O(T)	14.2 GiB	0.11 tok/s	1.0x (9.1 s/tok)
7.62B MoS + SCache (W=256, S=16)	**6.67**	**2.99 GiB MLP (3.72 GiB)**	**22.0 MiB O(1)**	**0 MB**	**24.7 tok/s**	**224.5x (62.7x INT4)**
72.71B Dense FP16 + Full KV	4.35	135.4 GiB (866 GiB OOM)	1,280 MiB O(T)	0 MB	Cluster Ref	Multi-A100 Only
72.71B AirLLM FP16 (NVMe* / Cloud)	4.35	<2 GiB (I/O-stalled)	1,280 MiB O(T)	135.4 GiB	0.026 / 0.0006	1.0x (38.7 s/tok NVMe)
72.71B MoS + StreamLLM (W=128)	6.37	38.12 GiB (31.4 GiB 8x)	40.0 MiB O(1)	0 MB (VRAM)	4.72 / 7.27	182x–280x (Amnesia)
72.71B MoS + SCache (W=256, S=16)	4.86	38.12 GiB (31.4 GiB 8x)	126.0 MiB O(1)	0 MB (VRAM)	3.90 / 7.27	150x–280x NVMe
72.71B MoS + SCache (W=384, S=16)	**4.75**	**38.12 GiB (31.4 GiB 8x)**	**166.0 MiB O(1)**	**0 MB (VRAM)**	**4.03 / 7.27**	**155x–280x (6,717x Cloud)**

Table 3 (Vector 7 — Multi-Scale 3D Pareto Frontier across 7.62B & 72.71B): *Throughput reports measured 2xT4 hybrid / pure VRAM tok/s at 72.71B. Canonical 20x4L shards (1.79 GiB each) also enable local generative execution on 8–16 GB Apple Silicon / consumer PCs.*

Adversarial Vector	Conventional Baseline	MoS / SCache O(1) Result	Audited 0.5B -> 7.62B -> 72.71B Finding
V2: Periodic Callback (W=64)	**72.7B Naive: 127.91; StreamLLM: 97.46**	**72.7B SCache Tail PPL: 19.51 (-86.8% KV)**	**5.00x lower tail PPL than StreamingLLM at 72.7B**
V3A: Contradictory Overwrite	Hebbian: cos=0.467, Ghost +0.501	Widrow-Hoff: cos=1.0000, Ghost -0.138	Zero ghosting in constant 34.8 KB bank
V3B: Passcode (445 tok, 72.7B)	**72.7B StreamingLLM NLL: 3.2414**	**72.7B SCache O(1) NLL: 0.5758**	**14.38x higher exact prob at 72.7B (zero tuning)**
V3B: Long Eviction (1.2k–1.7k)	StreamLLM: 2.961 (72B) / 3.683 (7B)	SCache: 1.279 (72B) / 1.667 (7B)	5.38x (72B, -83.4% KV) & 7.51x (7B, 1,745t)
V4: 500x DoS + Antipodal	Linear subspace drift / corruption	Target cos: 1.0000; Leak: 0.00000000	Exact ReLU cone immunity to hostile noise
V5: Full-Stack 72.7B QA/Code	**72.7B Dense+Full KV: 3/6, NLL 2.486**	**72.7B MoS+SCache: 3/6, Ans NLL 1.858**	**-72.7% MLP & -86.8% KV, lower answer NLL**
V6: Canonical 72.7B / 123B	72.7B: 135.4 GiB; 123B: 246 GB	72.7B: 20x1.79 GiB (<1.85 GiB active)	Bit-exact 4L vs 8L; 123B at 0.04 GiB post-group

Table 4 (Vectors 2–6 — Colossal 72.71B Long-Horizon Recall, Anti-Ghosting, DoS Immunity, and Group Streaming).

4. Boundary Conditions, Local 72B CLI Deployment & Reproducibility

(1) **Cross-Layer SwiGLU Permutation Sensitivity:** Sharing one dense base across *distinct* pretrained layers without permutation matching incurs distortion (KL ≈ 0.78); off-the-shelf models pair per-layer INT4 bases with ASVD scars (38.12 GiB at 72.71B), reserving single-base sharing for multi-tenant banks (-96.37% VRAM across 8 specialists at 72.71B). (2) **Associative Slot Capacity:** Fixed $S_{slots} \times d_h$ fast weights recover 5.38x–22.58x higher evicted passcode probability zero-shot; arbitrary long random string reproduction scales with S_{slots}. (3) **Open 72.7B Execution (mos-run):** Canonical 20×4-layer Qwen2.5-72B-Instruct MoS v2 shards (<1.80 GiB each), CLI (python -m scar_cache.cli_72b), 39/39 pytest suite, LaTeX source, and raw 0.5B/7.62B/72.71B JSON telemetry are archived at **DOI: 10.5281/zenodo.24143839** [13] and **github.com/byte-od/mixture-of-scars**.

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